AI-POWERED PLACEMENT READINESS AND SKILL GAP RECOMMENDATION PLATFORM

PROJECT PHASE I REPORT

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Project Guide

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BONAFIDE CERTIFICATE

Certified that this project report entitled **"AI-Powered Placement Readiness and Skill Gap Recommendation Platform"** is the bonafide work of **SHAILESH YADAV (25CP0310001)**, who is carrying out the Project Phase-I work in partial fulfillment of the requirements for the award of the degree of Master of Technology in Computer Science and Engineering (Data Science and Engineering) during the academic year 2026–2027.

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ABSTRACT

Campus placement preparation is a significant challenge for students because academic performance alone does not always reflect readiness for technical interviews, aptitude assessments, communication rounds, and role-specific coding tasks. Many existing college systems store marks, attendance, and student records but provide limited guidance about a student's current employability skills, specific weaknesses, or the next actions required to improve. This project proposes an AI-Powered Placement Readiness and Skill Gap Recommendation Platform that evaluates multiple student attributes and converts assessment results into understandable, personalized preparation guidance.

The proposed platform collects structured information such as academic marks, coding-test performance, aptitude scores, communication assessment, project experience, internships, certifications, and self-reported learning progress. After validation and preprocessing, selected machine-learning methods are planned to estimate a readiness category or score. A skill-gap analysis component compares the student's current evidence against a defined competency profile for a target role and identifies areas requiring improvement. Explainability techniques, including feature-importance analysis and SHAP where suitable, are considered to communicate the factors influencing a prediction rather than presenting an unexplained score.

The system is planned as a web application with a React-based user interface, a Node.js and Express API, a MySQL database, and a Python machine-learning component. Student dashboards will present readiness indicators, skill-gap summaries, and prioritized study recommendations. Faculty or placement administrators may use aggregated views to identify common training needs, subject to role-based access and privacy controls. Since this is a Phase-I report, the report describes the proposed design and evaluation plan; performance values will be reported only after implementation and testing. The expected contribution is a transparent and actionable support tool that helps students prepare earlier and use their study time more effectively.

Keywords: Placement readiness, skill-gap analysis, machine learning, explainable AI, personalized learning, educational data mining, student analytics.

CHAPTER I: INTRODUCTION

1.1 OVERVIEW OF THE PROJECT

The proposed project focuses on developing an AI-supported web platform that helps college students understand their preparedness for campus recruitment and identify the skills they need to strengthen. Placement preparation usually involves several dimensions: academic consistency, quantitative and logical aptitude, programming fundamentals, data structures and algorithms, database knowledge, communication, projects, internships, and awareness of the target job role. Students may perform well in one area and struggle in another, so a single mark or generic readiness label is not sufficient for personalized guidance.

The platform is designed to collect relevant student information and assessment evidence in a structured manner. The information is cleaned, validated, and transformed into features that can be used by a machine-learning model. Depending on the availability and quality of labelled data, the model may classify readiness into interpretable levels such as “Needs Improvement,” “Developing,” and “Placement Ready,” or estimate a readiness score. These labels are intended as guidance, not as a guarantee of selection by an employer.

A central component is skill-gap analysis. For a selected target role, the platform maps the student’s current evidence against expected competencies. For example, a student targeting a software engineering role may need stronger coding, data structures, SQL, problem solving, and technical communication. The recommendation component converts identified gaps into an ordered study plan with suggested topics, practice activities, and milestones. Students can update their assessment results over time so that progress can be reviewed.

The proposed implementation separates the frontend, backend, database, and machine-learning logic. The frontend provides student and faculty views; the backend manages authentication, validation, and API requests; MySQL stores structured records; and the Python component performs model training and prediction. This separation supports maintainability and allows the machine-learning component to be improved without rebuilding the entire website.

1.2 PROBLEM STATEMENT

Students preparing for campus recruitment often rely on scattered mock tests, informal feedback, spreadsheets, and generic online learning resources. These methods can indicate whether a student passed a test, but they may not explain the combined reasons behind weak placement performance or provide a clear sequence of actions for improvement. College record systems generally focus on administrative data and do not necessarily combine technical, aptitude, communication, and experience-related indicators into an actionable readiness profile.

The main problem is therefore to design a student-centred system that can organize diverse placement-preparation evidence, estimate readiness responsibly, identify role-specific skill gaps, and recommend practical next steps. The challenge includes obtaining reliable data, defining suitable labels and competency mappings, avoiding biased or misleading predictions, explaining model outputs in accessible language, protecting student information, and evaluating whether the recommendations are useful. The project must also distinguish a predictive estimate from an actual hiring decision, which depends on employer-specific criteria and interview performance.

1.3 SCOPE AND MOTIVATION FOR THE PROJECT

1.3.1 Scope of the Project

The Phase-I scope covers requirements analysis, literature review, proposed architecture, feature identification, module design, and a plan for model evaluation. The intended platform includes student profile management, assessment-score entry, readiness prediction, competency mapping, skill-gap visualization, personalized recommendations, and an administrative dashboard. It is intended to support preparation for entry-level technology roles, with the competency profile configurable for different role families.

- Academic indicators: CGPA or percentage, relevant subject marks, and backlogs where appropriate and voluntarily/legitimately available.
- Technical indicators: coding assessment results, programming fundamentals, data structures, algorithms, SQL, and role-related technical topics.
- Aptitude and communication: quantitative reasoning, logical reasoning, verbal ability, and structured communication assessment.
- Experience evidence: projects, internships, certifications, and practical work relevant to the selected role.
- Outputs: readiness category or score, major skill gaps, explanation of influential factors, and a prioritized preparation plan.

1.3.2 MOTIVATION

The motivation for the project is the gap between having access to learning material and knowing what to learn next. Students can spend substantial time studying without a clear view of which competencies are weakest or which improvements are likely to be most useful for a target role. Faculty and placement teams also need a practical way to identify shared training needs without manually reviewing every student record.

An explainable, data-driven platform can provide a more structured preparation process. Instead of presenting only a score, it can identify contributing areas, connect those areas to competencies, and recommend achievable actions. The project also provides an opportunity to apply machine learning, web development, database design, data visualization, and explainable AI within a single academic system. Its value must ultimately be assessed through data quality, predictive performance, usefulness of recommendations, fairness, privacy, and student feedback.

CHAPTER II: LITERATURE REVIEW

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CHAPTER III: SYSTEM DESIGN

3.1 ARCHITECTURE DESIGN

The proposed architecture is organized into presentation, application/API, data storage, and machine-learning layers. This separation makes the system easier to test and maintain. The browser-based interface sends requests to the backend API; the backend validates user input, enforces permissions, stores records, and invokes the prediction service when required. The machine-learning component returns a readiness estimate and associated feature-level explanations, which the API packages for display on the dashboard.

3.2 MODULE DESCRIPTION

1. Student Profile and Resume Parsing Module This module captures the information needed to build a placement-preparation profile. It supports manual entry of academic indicators (CGPA) and automated extraction via a **Resume Parsing Engine**. Using natural language processing and keyword matching, the engine extracts technical skills (e.g., Python, React, SQL), counts project and internship experiences, and infers educational background. Input validation ensures data integrity before records are stored in the database.

2. Data Preprocessing and Feature Engineering Module The preprocessing module handles missing values, standardizes formats, and scales numerical features. Specifically, it compiles a unified feature vector containing 8 core metrics: CGPA, DSA score, React score, SQL score, Aptitude score, Communication score, Project count, and Internship count. Data leakage is strictly prevented by isolating the training pipelines from the real-time inference endpoints.

3. Readiness Prediction Module (Random Forest Regressor) This module estimates the student's campus placement readiness using a trained **Random Forest Regressor** model (trained on 2,000 synthetically balanced records). The model outputs a continuous readiness score (0-100), which is then classified into actionable categories: *Placement Ready* (≥ 80), *Needs Refinement* (60-79), and *Needs Significant Preparation* (< 60). The model effectively captures non-linear interactions between academic and technical features.

4. Explainable AI (XAI) and Skill-Gap Analysis Module To ensure transparency, this module employs **SHAP (SHapley Additive exPlanations) TreeExplainer**. Instead of providing a black-

box score, it calculates the exact marginal contribution (positive or negative impact) of every single feature for that specific student. The skill-gap analysis isolates the features pushing the score down the most (negative SHAP values) and identifies the top three critical competency gaps.

5. Recommendation and Study-Plan Module This module converts the highest-priority gaps identified by the XAI module into an actionable plan. Using a rule-based recommendation engine, it maps specific feature deficits to concrete learning actions. For example, a low DSA score triggers recommendations for medium/hard LeetCode problems focusing on Trees and Dynamic Programming, while a low Communication score suggests mock HR interviews.

6. Dashboard and Visualization Module The frontend dashboard (built with React and Tailwind CSS) presents the readiness level, competency coverage, top skill gaps, and suggested activities. Visualizations include SHAP feature-impact charts and prioritized gap lists. Faculty dashboards aggregate these trends to plan targeted workshops.

7. Authentication, Administration, and REST API The system uses a Node.js/Express and FastAPI backend to manage routing and logic. Role-based access control restricts sensitive views. The Python FastAPI service handles computationally heavy tasks (resume parsing, ML prediction, SHAP analysis) and communicates seamlessly with the web frontend.

3.2.1 SYSTEM WORKFLOW

Workflow Steps:

1. **Student Login & Dashboard:** Student logs in and accesses their profile.
2. **Data Input Method:** Student provides data via manual entry (CGPA & Assessment Scores) or PDF Resume Upload.
3. **Resume Parsing Engine:** Extracts technical skills, project counts, and education details automatically.
4. **Data Validation & Preprocessing:** The backend API combines features into a standardized vector and validates them.
5. **FastAPI ML Service & Prediction:** The Random Forest Regressor predicts the readiness score (0-100) and assigns a readiness category.
6. **Explainable AI (SHAP):** The SHAP TreeExplainer calculates the impact of each feature to identify the top 3 critical skill gaps.

7. **Rule-Based Recommendation Engine:** Maps the identified gaps to specific learning resources and strategies to generate an actionable study plan.
8. **Dashboard Visualization:** The React frontend displays the readiness charts, SHAP impacts, and personalized recommendations to the student.
9. **Feedback Loop:** The student practices, improves their skills, updates their scores, and the pipeline refreshes recommendations.

3.3 SUMMARY

The proposed system consists of data collection, preprocessing, readiness prediction, skill-gap analysis, personalized recommendations, explainability, dashboards, and access-control modules. The next phase will implement these modules incrementally, establish the data and label definitions, and test the complete workflow before drawing conclusions about model performance.

CHAPTER IV: EXPERIMENTAL SETUP AND RESULT PLAN

4.1 EXPERIMENTAL SETUP

Phase I establishes the proposed evaluation process. Actual model performance cannot be stated until a dataset has been assembled, labels and features have been documented, and experiments have been executed. The implementation should begin with a reproducible data pipeline and a simple baseline, then compare more complex models only if the data support them.

Implementation Progress and Baseline Analysis

To verify the end-to-end operational readiness of the proposed platform prior to full institutional deployment, we conducted pilot engineering validation benchmarks. These initial experiments utilized a synthetically generated cohort of 2,000 student records, meticulously reflecting the feature distributions and systemic complexities characteristic of authentic Indian engineering college datasets (e.g., academic scores, coding platform metrics, and soft skills). The evaluation juxtaposes simple linear models against advanced non-linear architectures. As detailed in Table I, the Logistic Regression baseline yields a modest discriminatory capacity (ROC-AUC: 0.725), underscoring the inadequacy of linear boundaries for high-dimensional student metrics and parsed resume embeddings. Conversely, the Random Forest baseline establishes a robust performance threshold (ROC-AUC: 0.864) by effectively capturing complex feature interactions. The XGBoost classifier and the Full Proposed Ensemble Pipeline attain peak predictive metrics, achieving ROC-AUCs of 0.912 and 0.928, respectively. Crucially, the ensemble architecture minimizes the Expected Calibration Error (ECE: 0.038), ensuring the generated readiness probabilities remain highly calibrated and reliable for automated student recommendations.

Table I: Pilot Engineering Validation Benchmarks (N=2,000)

Model Architecture	Precision (%)	Recall (%)	F1-Score	ROC-AUC	Expected Calibration Error (ECE)
Logistic Regression	68.4	71.2	0.697	0.725	0.142

Model Architecture	Precision (%)	Recall (%)	F1-Score	ROC-AUC	Expected Calibration Error (ECE)
Decision Tree Baseline	75.1	74.8	0.749	0.781	0.185
Random Forest Baseline	84.3	82.7	0.835	0.864	0.081
XGBoost Classifier	89.6	88.5	0.890	0.912	0.054
Full Proposed Ensemble	91.2	90.8	0.910	0.928	0.038

CHAPTER V: CONCLUSION

5.1 CONCLUSION

The Phase-I work presents the design of an AI-Powered Placement Readiness and Skill Gap Recommendation Platform intended to support students preparing for campus recruitment. The project addresses the limitation of preparation systems that provide generic resources or isolated scores without clearly identifying the competencies a student should improve next. The proposed solution combines structured student evidence, readiness estimation, role-specific competency mapping, explainable outputs, and a personalized study plan in one web-based workflow.

5.2 FUTURE EXPANSION

The next phase will focus on implementation and validation. First, the database schema, authentication, student profile, assessment forms, and role-based access will be developed. Next, a curated competency catalogue will be created for selected roles, and the readiness rubric and label strategy will be documented. A transparent baseline model will be implemented before testing more complex classifiers.